

Educational Walkthrough: TCP Flood and Port Scan Detector

Introduction

This walkthrough provides a detailed educational explanation of the concepts and components involved in the TCP Flood and Port Scan Detector project. It is intended for students, instructors, and cybersecurity learners who want to understand not just how the tool works, but why it works—by mapping network security theory to practical detection logic and hands-on software interaction.

1. Networking Concepts Behind the Tool

1.1 The TCP/IP Model

The tool operates primarily at the transport layer of the TCP/IP model, which is responsible for reliable communication between devices. TCP (Transmission Control Protocol) manages packet delivery using mechanisms such as sequence numbers, acknowledgments, and flags.

1.2 The TCP Three-Way Handshake

The three-way handshake is the foundational process by which a TCP connection is established:

1. SYN – The client sends a synchronization packet to start a connection.
2. SYN-ACK – The server responds, acknowledging the SYN and offering to synchronize.
3. ACK – The client replies, completing the connection.

SYN flood attacks disrupt this process by sending many SYN packets but never replying to the server's SYN-ACK. This leaves the server waiting and ties up resources.

1.3 Port Scanning

Port scans are used by attackers to identify open or vulnerable services running on a target. This involves systematically sending packets to different port numbers and recording responses.

If a single IP probes many unique ports in a short time, it's often a sign of reconnaissance—an early step in the attack chain.

1.4 TCP Flags and Detection

Every TCP packet contains flags that describe its purpose. Key flags used in this project include:

- SYN: Connection initiation.
- ACK: Acknowledgment of receipt.
- RST: Reset the connection.

By tracking SYN flags over time and counting how many different ports are contacted, the tool determines whether an attack is underway.

2. Applying Theory to Practice: How the Tool Works

The TCP Flood and Port Scan Detector was built using Scapy for packet capture and PyQt5 for the graphical interface. It displays packets live, interprets their flags, and alerts users to anomalies.

Each design decision in the code reflects a theoretical concept from the sections above.

2.1 Detection Logic

- SYN Flood: If more than 100 SYN packets are received from a single source within 10 seconds, an alert is triggered.
 - Port Scan: If a source IP sends packets to 10 or more unique ports, an alert is issued.
- These thresholds can be adjusted but are based on typical intrusion detection guidelines.

2.2 User Interface as a Learning Tool

- The packet table displays time, source/destination IPs and ports, and flags.
- Row colors correspond to TCP flags (yellow=SYN, red=RST, blue=ACK, bright red=attacks).
- A graph tracks packets per second to visualize traffic surges.
- Tooltip headers explain each field in plain language.
- A detail panel shows full packet content when clicked.

2.3 Working with PCAP Files

The tool also allows users to load and analyze `.pcap` files created with Wireshark or tcpdump. This enables offline experimentation and traffic analysis, making it a flexible learning tool.

3. Educational Use Cases

3.1 For Students

- Visualize what happens during a port scan or SYN flood.
- Learn to interpret TCP flags in a GUI, not just Wireshark.
- Use filters to isolate specific traffic types (e.g., all SYNs from one IP).
- Load real or simulated `.pcap` files and observe traffic patterns.

3.2 For Instructors

- Create simulation labs using Nmap and hping3.
- Ask students to explain why alerts triggered based on packet behavior.
- Use `.pcap` files from real incidents for forensic analysis exercises.
- Demonstrate DoS attack mechanics safely and visually.

4. Hands-On Lab Simulation Guide

To simulate attacks and use the tool in a lab setting, follow these instructions:

1. Launch the tool and start capture.

2. From another terminal or device, run:
 - Port scan: ``nmap -p 1-100 <target_ip>``
 - SYN flood: ``hping3 --flood -S -p 80 <target_ip>``
3. Observe color-coded rows, graph spikes, and pop-up alerts.
4. Pause capture, review suspicious packets in detail view.
5. Export to ``pcap`` and load in Wireshark for comparison.

5. SYN Floods and Port Scans: Why They Matter

5.1 Why SYN Floods Consume Resources

SYN floods overwhelm servers by sending massive amounts of TCP SYN packets, leaving the server with many half-open connections.

Each pending connection occupies space in the server's backlog queue, consuming memory and CPU resources.

Because the attacker never completes the TCP handshake, the server waits for a timeout (typically 30–60 seconds).

If the backlog fills up, legitimate users cannot connect.

5.2 How Port Scans Work

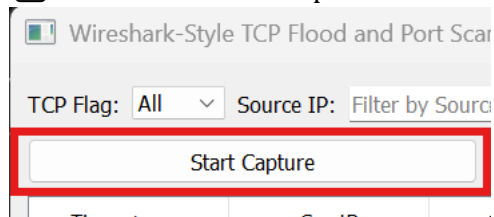
A port scan systematically sends TCP SYN packets to various ports on a target to discover open services.

The tool detects a port scan by tracking how many different destination ports are probed by a single source IP within a short timeframe.

6. Expanded Lab Simulation Guide

6.1 Step-by-Step Instructions

- 1 Start the Detector: Open the tool and begin real-time capture.



- 2 Simulate Port Scan: Run: `nmap -p 1-100 <target_ip>`

- Observe the GUI table and graph.

- Guiding Questions:

- What TCP flags do you see?

- How does the detector highlight the scan?

3 Simulate SYN Flood: Run: `hping3 --flood -S -p 80 <target_ip>`

- Watch for packet spikes and alert pop-ups.
- Guiding Questions:
 - How quickly does the flood trigger an alert?
 - What happens to the graph's spike rate?

4 Pause & Review: Examine packet details using the GUI's packet viewer.

5 Bonus: Export to .pcap and inspect in Wireshark for deeper analysis.

6.2 Expected Observations

- Port scan: Rapid sequence of SYN packets across many ports.
- SYN flood: Consistent high-rate SYN traffic to one port.
- Graph: Spikes in packets/sec during attacks.
- Alerts: Port scan and flood alerts trigger based on thresholds.

7. Diagrams (Recommended for Inclusion)

- TCP Handshake Diagram: Shows SYN, SYN-ACK, ACK steps.
- SYN Flood Diagram: Visual of repeated SYNs with no ACKs, backlog overflow.
- Port Scan Diagram: One source probing multiple ports.

8. Reflection and Learning Outcomes

Using this tool, learners will gain insight into how real-time monitoring systems detect threats. They'll better understand the structure and purpose of TCP packets, the difference between normal and suspicious traffic, and how security tools translate raw data into actionable alerts. The hands-on, visual nature of the interface supports retention and active learning.

Conclusion

This tool brings theoretical network security knowledge to life. It is more than code—it is a teaching platform, a diagnostic interface, and a simulation engine all in one.

By blending real-time monitoring with educational design, it helps users understand not just what is happening, but why it matters.